# Akshay Kanade


## CONTACT

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**(Home)**

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## SKILLS

- Programming: Python, C++, SQL, C
- Machine Learning & AI: Deep Learning, ViTs Transfer Learning, Supervised & Unsupervised Learning, Pruning, Quantization
- Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, Keras, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Detectron2, ONNX
- MLOps & Deployment: MLflow, DVC, Docker, Git, CI/CD
- Data Annotation & tooling: CVAT, Label Studio, Dataset Curation, Augmentation, Quality Validation
- Systems & Tools: Linux, ROS, Jupyter, VS Code

## WORK EXPERIENCE

**Software Developer (AI/ML Engineer) |** Starting Jan 2024
OneVision Software AG, Regensburg, Germany

- Architected end-to-end perception pipelines from data ingestion to production deployment, reducing delivery cycles by 50%
- Accelerated inference workloads by 70% through TensorRT optimization, ONNX conversion, and mixed-precision execution
- Evaluated CNN, ViT, and foundation models to guide architecture selection for diverse industrial vision tasks.
- Integrated explainability and validation techniques to improve model robustness and stakeholder trust.
- Coordinated with platform, DevOps, and product teams using CI/CD-driven MLOps workflows.

**Working Student & Master Thesis (Computer Vision & ML Engineering)** | Oct 2022 to Dec 2023
AVL Software and Functions GmbH, Regensburg, Germany

- Engineered containerized object detection pipelines on ARM devices, cutting inference latency by 50%.
- Compressed deep learning models using pruning and quantization for edge deployment.
- Automated ROS bag processing and annotation workflows using Python and SQL.
- Validated LiDAR and radar sensor outputs to ensure perception accuracy in ADAS systems.

**Data analyst |** Jan 2019 to Aug 2021
Accenture, Mumbai, India

- Designed scalable ETL pipelines improving data throughput by 65%
- Analyzed large relational datasets using SQL to support enterprise reporting
- Documented data workflows and quality checks in regulated environments

## EDUCATION AND TRAINING

**Master of Engineering: Electrical Engineering and Embedded Systems |** Sep 2021 to Oct 2023
Ravensburg Weingarten University, Germany

Lidar and Radar Systems, Computer Vision, Embedded Computing, Advanced Control Systems, Autonomous Driving, Signal Processing.

**Bachelor of Engineering: Electronics and Telecommunication |** May 2014 to June 2018
Savitribai Phule Pune University, India

Electronics, Embedded Processors, Microcontroller and application, Computer Organization

## LANGUAGES

German: A2, learning B1;
English: Fluent;
Hindi: Native language;
Marathi: Mother Tongue

## PUBLICATION

Evaluation of dimensions of a vehicle using Velodyne and Blickfeld LiDAR

https://www.ijser.org/onlineResearchPaperViewer.aspx?Evaluation-of-dimensions-of-a-vehicle-using-Velodyne-and-Blickfeld-LiDAR.pdf

International Journal of Scientific & Engineering Research Volume 12, Issue 12, December-2021 ISSN 2229-5518

## HOBBIES

Cricket (**Captain** Regensburg Team), Volunteer Work, Trekking

## German Driving License

(Class B)

## PROJECTS

**Computer Vision Object Detection & Segmentation Pipeline | **2024

- Developed and deployed YOLOv5, Faster R-CNN, and U-Net for detection and segmentation tasks.
- Automated data preprocessing and augmentation using OpenCV and Albumentations.
- Optimized for edge devices (Jetson Nano, Raspberry Pi) with pruning, quantization, and TensorRT for 60% faster inference.

**MLOps Deployment & Monitoring |** 2024

- Built CI/CD pipelines with MLflow, DVC, and Docker for versioning and scalable model deployment.
- Developed interactive dashboards with Power BI and Plotly for real-time model performance tracking.

**Camera Calibration & Image Processing Automation |** 2022

- Performed intrinsic camera calibration using OpenCV, achieving a reprojection error of 0.0156%.
- Implemented edge detection, perspective transformation, and geometric corrections to support automated vision systems.
- Validated calibration accuracy for downstream perception tasks.

**Vehicle Position & Dimension Prediction |** Nov 2021

- Fused Blickfeld and Velodyne LiDAR data to estimate vehicle positions and dimensions.
- Applied regression models to forecast future vehicle positions accurately.

## CERTIFICATES

**DevOps Fundamental CI/CD with AWS + Docker + Ansible + Jenkins**

CI/CD automation using AWS, Docker containers, Ansible configuration, Jenkins pipelines, and Git version control.

**Python for Data Science and Machine Learning Bootcamp**

Data analysis and model building using NumPy, Pandas, Seaborn, Matplotlib, Plotly, and Scikit-Learn.

**TensorFlow Developer Certificate**

Deep learning specialization covering CNNs, RNNs, and ANN models for computer vision and NLP applications.

**Machine Learning, Data Science and Deep Learning with Python**

Comprehensive ML pipeline development, feature engineering, model training, and deployment using Python.

**MLOps with MLflow, Docker, and Kubernetes for Deployment**

End-to-end ML lifecycle management with model tracking, CI/CD, and scalable deployment.

**Computer Vision A–Z™: Learn OpenCV, SSD, YOLO, GANs**

Object detection, image segmentation, and generative vision networks for real-world computer vision applications.